SCT3258 FAQ

Confidential，样
Internal Use Only

Version: 1.0

Date: 2013/11/04

Copy Rights © 2013, Wuxi Sicomm Communication Technologies, Inc

1 Protocol

1. Does SCT3258 support the Europe standard DPMR (ETSI TS 102 490)?
   SCT3258 supports the entire function in DPMR (ETSI TS 102 490) including voice call, slow data, Type 1 data and Type 2 data. SCT3258 doesn't support Type 3 (Packet) data.

2. Does SCT3258 support the Europe standard DPMR (ETSI TS 102 658)?
   SCT3258 supports Mode 1 and Mode 2, but doesn't support Mode 3.

3. Does SCT3258 support Europe DMR standard (ETSI TS 102 361).?
   Yes. SCT3258 support ETSI TS 102 361

4. Does SCT3258 support the Japanese standard DCR?
   SCT3258 supports the Japanese standard DCR on physical layer. But the user need develop the Layer 2 software.

5. Does SCT3258 support the American standard NXDN?
   Yes.SCT3258 supports the American standard NXDN on physical layer. But the user need develop the Layer 2 software.

6. Does SCT3258 support Chinese CDMR standard?
   Yes. SCT3258 support CDMR Tier 1.

7. Does SCT3258 support analog voice call?
   SCT3258 supports analog voice calls. It also supports CTCSS and DCS generation and detection.

8. Does SCT3258 support automatic detection of analog and DPMR calls?
   If SCT3258 works with conventional RF circuits, it supports automatic detection of analog and DPMR calls. If SCT3258 works with SCT3700, it does not support automatic detection of analog and DPMR calls.

9. Does SCT3258 support automatic detection of analog and DMR calls?
   No, SCT3258 does not support automatic detection of analog and DMR calls.

10. Does SCT3258 support automatic detection of DPMR and DMR calls?
    No, SCT3258 does not support automatic detection of DPMR and DMR calls.

Confidential，供内部使用 Internal Use Only

Copy Rights © 2013, Wuxi Sicomm Communication Technologies, Inc

2 Hardware

1. Does the MCU take part in the controlling of codec or the SCT3258 deal with the data of codec automatically?

   The codec is controlled by SCT3258 while the MCU only communicates with SCT3258.

2. How can I configure the PIN HSTS of HPI?

   When HSTS is asserted HIGH, the HPI reads the contents of the HPI status register. For normal use, this signal should be tied GROUND.

3. How can I configure the PIO[3:0] ports?

   a) PIO0: SCK, clock signal for I2C interface; or MOSI, data output line for SPI interface.
   b) PIO1: SDA, data line for I2C interface; or clock signal for SPI interface ; also as mode PIN for HPI mode select during reset: 0 for Intel Mode and 1 for Motorola Mode
   c) PIO2:chip-select for SPI, CE
   d) PIO3: In DMR mode, it outputs 60 ms slot timing, aligned with transmitted or received TDMA signals. Active High. In DPMR mode, PIO3 outputs 1 Hz square wave if SCT3258 is functionally normally. If PIO3 outputs a constant level, the ST3258 need to be reset.

4. What are the requirements to power on the SCT3258?

   For power-on timing requirements for the SCT3258, refer to Section 12.2, “Reset and Power-on Timing” in 3258_datasheet. The SCT3258 requires a 1.2 V supply for the DSP core and a 2.5-3.3 V supply for the I/O ports. The DSP core powers supply first，I/O power supply after. Both 3.3v and 1.2v have to power down as soon as possible. Otherwise, SCT3258 may reset incompletely and may not work normally when the user power on again after power on and power down fast.

5. What do I need to know about resetting the SCT3258?

   Please pay attention to the PIO1’s level. If the HPI port is configured as Intel Mode, the PIO1 should be LOW. If the HPI port is configured as Motorola Mode, the PIO1 should be HIGH. There is a pull-up resistor in PIO1.If selecting Motorola Mode, we can power on SCT3258 without setting PIO1. If selecting Intel Mode, please make sure the PIO1 is LOW. Generally, we can use a reset delay circuit or make PIO3 to be controlled by the MCU.
   The level in RSTN should hold LOW for at least 500 ms. During the reset, the level of PLLSEL2 and PLLBYPASS level should remain intact.

6. When can I change SCT3258 System clock frequency?

Confidential，Internal Use Only

The System clock frequency can be changed by changing the level of PLLSEL2 or PLLBYPASS. To avoid glitches, it is recommended that you change the system clock frequency while SCT3258 is in SLEEP or HALT state. You can put SCT3258 in SLEEP or HALT state with Low Power command (field ID 0x15). After you change the system clock frequency, you can bring SCT3258 back to normal working state from SLEEP state with INT0. You can bring SCT3258 back to normal working state from HALT state with NMI.

7. Which pin should I connect to the MCU ?

The pins in the table should be connected to the MCU.

Pin Name	Pin Number	Must（M） Optional（O）	Description
HCSN	63	M	HPI Select and Enable Signal
HRDN	61	M	HPI Read Signal
HWRN	62	M	HPI Write Signal
HOBIB	60	M	HPI Status Signal
HPIDATA[7:0]	50-57	M	HPI Data Signal
RSTN	59	M	Reset Signal
NMI	58	O	Software Reset Signal , without re-downloading the firmware.
INT0	64	M	Interrupt Signal
PLLBYPASS	45	O	PLL Bypass Signal
PLLSEL2	48	O	PLL Frequency Select Signal
PIO3/RF_TIMING	15	O	SCT3258 RF Timing Signal in DMR mode, or work status signal

8. What is the requirements of SCT3258 clock input?

SCT3258 requires a fixed clock input of 12.288 MHz, with a peak to peak voltage of 1.2 V to 3.3 V.

9. What are the requirements of the external codec? If I choose the codec from AKM or ADI, can SCT3258 control the codec's registers?

Other codecs can be used with SCT3258, as long as it can support I2C interface for control, 24kHz sampling rate and the serial interface like DSP mode of TI. Details can be found in "SCT3258_Datasheet"

10. Does SCT3258 support two point modulation?

Confidential 试样 Internal Use Only

Copy Rights © 2013, Wuxi Sicomm Communication Technologies, Inc

Yes. SCT3258 support two point modulation. 3258 has two outputs in TX mode, connecting to VCO and VCTCXO port independently to facilitate 2 point modulation. The user can also send command to codec via SCT3258 to adjust the amplitude of the two outputs.

11. Does SCT3258 support I/Q modulation?

Yes.SCT3258 supports zero-IF or low-IF (0-3.125kHz) in receive mode and zeros-IF in transmit mode.

12. In SCT3258 Application Circuits, what is the function of the delay circuit of PIO1? Can I leave out this?

The HPI port can be configured as Intel Mode or Motorola Mode. When SCT3258 reset, it will detect the level of PIO1. HIGN is Motorola Mode and LOW is Intel Mode. The delay circuit ensures the level is LOW to be Intel Mode. If the user chooses Motorola Mode, the circuit can be left out and make the PIO1 unconnected.

13. In analog mode, are there two independent registers to control the CDCSS/CTCSS Frequency Deviation and voice in SCT3258?

Yes. There are two independent registers to control the CDCSS/CTCSS Frequency Deviation

14. Does the TX modulation has limiting function?

Yes.

15. What is the functions and initial conditions of the PIN PLL-SEL, NIMI, INTO, BYPASS, RST in SCT3258?

PLL-SEL: control the frequency division factor of PLL in SCT3258. The PLL frequency is 110.592MHz when PLLSEL is high and 61.44MHz when PLLSEL is low.

NMI: Active HIGH. Reset without re-downloading the code from FLASH.

INT0: Active HIGH. Wake up SCT3258 in the sleep mode.

BYPASS: Active HIGH. Use XO clock instead of PLL clock to reduce the power consumption. The system CLOCK is 12.288MHz of XO when PLLBYPASS is high. The system clock is PLL clock when PLLBYPASS is LOW.

RSTN: RSTN must be asserted (LOW) for a period of at least 500 ms then set HIGH.

The user can configure the pins by himself when initial SCT3258.

Confidential，限制
Internal Use Only

Copy Rights © 2013, Wuxi Sicomm Communication Technologies, Inc

16. How is the power consumption of SCT3258?

SCT3258 uses dual power supply and the work current is related with work frequency. When the frequency is 110.592MHz (PLLSEL=1), the current of 1.2v is 37mA and 3.3v is 5mA.When the frequency is 61.44MHz (PLLSEL=0), the current of 1.2v is 23mA and 3.3v is 3mA.We recommend using of the low frequency (PLLSEL = 0) when the radio work in half-duplex DPMR mode (TX or RX), and high frequency (PLLSEL = 1) when the radio work in DMR mode.

We recommend set the PLL-SEL HIGH during bootloading to start more quickly. After bootload is successful, set the PLL-SEL LOW in DPMR mode to save power.

17. If I use the other codec instead of the codec your recommend, what should I notice?

Please read the SCT3258_DATASHEET carefully and pay special attention to the following:.

1. 16-bit, stereo mode
2. MSB first
3. Left channel first, right channel after
4. The sampling rate is 24K
5. I2C address is 8bit. If it is 7bit, please shift lift 1bit to be 8bit

18. Does SCT3258 support DC coupling?

Both the FM demodulated input and modulated output support DC coupling. The user can use the DC_OFFSET command to adjust the DC offsets.

19. Does SCT3258 support VOX detection?

SCT3258 support the VOX detection indirectly. MCU sends the VOX_ENERGY command to get the current voice energy as the base for VOX detection.

Confidential，供样
Internal Use Only

Copy Rights © 2013, Wuxi Sicomm Communication Technologies, Inc

3 Packet Interface

1. How can I verify the process of downloading the bootloader is correct?

   Download testport3.dat, the easiest bootlader.If it is correct, you will see a square wave on PIO3 with T = 0.5s and a square wave on PIO2 with T = 1s.

2. I download the testport3.dat but there is no expected result. What should I do?

   Please check the following:
   a. The timing of HPI.
   b. MSB followed by LSB.
   c. PIO1 is low at reset for Intel Mode and high for Motorola mode

3. How do I boot SCT3258 as in DPMR mode, or DMR mode?
   SCT3258 can be booted up as a DPMR / Analog processor, or a DMR / Analog processor. The 1024 word boot loader is different for DPMR or DMR. The boot loader for different mode can be obtained from Sicomm Support.

4. How can I know the DPMR firmware is successfully booted?

   After DPMR firmware is successfully booted, the user can observe the following:
   a. PIO3 sends out a 1 Hz square wave signal
   b. SCT3258 sends an ACK_MESSAGE to the host through HPI, with 17 01

5. How can I know the DMR firmware is successfully booted?
   After DMR firmware is successfully booted, the user can observe the following:
   a. SCT3258 sends an ACK_MESSAGE to the host through HPI, with 17 01

6. How long can I send command to SCT3258 after bootloading is successful?

   After SCT3258 send out an ACK_MESSAGE with 17 01, the MCU can send message to SCT3258 after 1 second.

7. Can I upgrade the Sicomm chip with SCT3258 client? For example, there is a bug in SCT3258F built-in software, how to update or upgrade the chip's software?

   Yes. With Sicomm provided PC software, the user can upgrade SCT3258 already installed on radios. The software sends new firmware to MCU through UART serial interface. The MCU on the radio relays the information to SCT3258 through HPI. The user should implement the UART – HPI relay function on the MCU when developing the products. Sicomm can provide

Confidential，样
Internal Use Only

example C code of the relay function.

8. Can the total number of bytes in a packet the MCU send to SCT3258 be odd?

   No. The total number of bytes in a packet the MCU send to SCT3258 should be even. If the total number of byte is odd, a zero is appended at the end. This appended zero will not be counted in the LENGTH field.

9. Is even or odd parity check used in SCT3258?

   Even.

10. Can the software in current version send and received the extended wake-up Header in DPMR mode?

    Yes. The user can set the number of wake-up Header with DPMR_CALL_OPTION. The receiver will send the wake-up Header first and then the normal Header.

11. How to test BER?

    SCT3258 has a mode to test BER especially. Details can be found in “SCT3258_ Packet Interface”.

12. How to calculate the parity byte?

    The second byte of the parity field is the parity byte. It is obtained by “Exclusive-oring” every byte in the packet, except for the START_BYTES and the parity byte (last byte), together.

    For example:
    Send Command: 84 A9 61 00 03 23 1E 2F 11
    00^03^23^1e^2f = 11

13. If the Called ID of transfer doesn’t match the Own ID of receiver, or the Color Code doesn’t match, can I make a call?

    We can set the CARRIER_READY command to decide whether to take the call or not when Called ID or Color Code does not match. When bit 2 is 1, SCT3258 does not take the call. When bit 2 is 0, SCT3258 will take the call and send the un-match message to MCU through CALL_MATCH. It is up to the MCU to decide whether to take the call or not.

14. I set the REPORT_FIELD to select the messages to report to MCU at receiver initiation. SCT3258 will report the message (such as DPMR_CALL_OPTION) at first call. Why the message is not reported at the second time?

Confidential，禁样
Internal Use Only

Please set the CARRIER_READY (19 01) again.

15. Since SCT3258 can write the register of TI codec, what about reading? If I can read, what should I do?

    Codec register is only readable.

16. We see the 4FSK signal at the HP_L port, which is the same as the signal on LO_R and LO_L port. Is this normal?

    The HP_L port and the LO_L port output the same signals. But HP_L port can be turned off by sending command to codec in TX mode.
    For example, for TI TLV320AIC3204, we can send 84 a9 61 00 06 40 00 01 09 0c 00 00 to turn off and 84 a9 61 00 06 40 00 01 09 2c 00 00 to turn on.

Confidential，祺祥
Internal Use Only

Copy Rights © 2013, Wuxi Sicomm Communication Technologies, Inc

4 Evaluation board and development board

1. How to test the power consumption?
   a) Please use the external 5V power supply that can display current. Please remove the jumper next to the power switch to avoid the conflict with USB power.
   b) There are three jumpers on SCT3258 development board. They are used to measure the current of 3.3V for codec, 3.3V for SCT3258 IO port, and 1.2V for SCT3258 core independently.
   c) Use CHIP_LOWPWR command (Field ID = 0x15) to change the work mode to measure the current of the components.
      0: Normal
      1: Stop TX and RX processing
      2: Reserved
      3: Power down certain peripheral of SCT3258 including serial port, DMA and timers.
      4: SCT3258 enters IDLE mode
      5: SCT3258 enters Sleep mode
      6: SCT3258 enters Halt mode

   In addition, we can set the value of bit 5 and bit 4 to shut down the vocoder and codec.

   We can get the power consumption by calculating the data.

2. I use DPMR Call Control Interface in the PC Host Controller to set Transmitter and Receiver's Call Mode to Voice+SlowData. One of the Evaluation boards is configured to transmit and the other Evaluation board is configured to receive. The receiver can receive the correct call ID and can communicate at the same time. But when I send SlowData I can not send more than two English or Chinese characters. If I use one board to loop back the transmit to the receive port, it is OK. Why is that?

   The "Receive Init" button on Transmitter and receiver need to pressed.

Confidential，祺祥
Internal Use Only

5 Overall Communication

1. Why can't I make a call when the settings are correct and the wave is good?

   When the settings are correct, please try to change the direction of CARRIER_READY in receiver. If we cannot set the receiver polarity, please set the MOD_GAIN (0x2c) to adjust the signal's polarity in transmitter. For example, change the MOD_GAIN from 0x0800 to 0xF800 will reverse the transmit polarity.. In addition, the Color Code of transmitter and receiver should match.

2. Why the receiver decodes very slowly?

   The decoding speed is related to the signal's quality. If the BER is high, please read the relative documents to decrease the BER to the minimum. With the limitation of RF, there may be an oblique wave at the beginning of the transmission. We can watch the waveform of the signal to SCT3258 to judge the details. We can add the length of preamble or a length of silence with DPMR_PREAMBLE_LENGTH command to make the receiver receive the header frame after the oblique wave.

3. When I use ANALOG_TONE to send tone to the far end in the call, can I make the speaker of CODEC output the tone and make TCXO keep outputting the far end signal?

   No. The codec can only be used in two channels at the same time. When we use ANALOG_TONE, it will affect the TCXO output definitely.

4. When I switch RX mode and TX mode continuously, I find that RX decode is slow or even failed.

   When the BER is low and the other settings are correct, please check whether the delay time between sending CALL_END and receiving 0x17 0x0A (the response after the baseband send the END frame) is too short. The longest time of waiting the response may be 340ms for DPMR, or 360 ms for DMR. So please make sure the waiting is correct, or it will affect the quality of the second call when you force to make the second call if the waiting time is less than 340ms.

Confidential 样稿
Internal Use Only

Copy Rights © 2013, Wuxi Sicomm Communication Technologies, Inc

6 DPMR Protocol Related

1. Can I set the CC (Color Code) arbitrarily? Can the CC of transmitter and receiver be different?

   No. When we set the CC, we must follow the DPMR protocol with different frequency match the different CC.
   If the CC of transmitter and receiver doesn't make, the receiver will report CALL_MATCH with CC not match.

2. Why do I receive an error response when I set the item EP in M_V_F_EP?

   For the item EP is only exist in ETSI TS 102 658, please set the PROTOCOL_OPTION command to support ETSI TS 102 658.

3. When I use DPMR_DATA_FRAME command to send DATA, why there is always a delay after Call start then the sending will be normal?

   There is no need to delay after CALL_START before sending data. To ensure the continuous between data part and header frame, DONOT delay after Call Start.
   The reasons for the abnormal delay could be:
   1. In TX, the correct response of DPMR_DATA_FRAME and DPMR_SLD should be 84 A9 61 00 01 20 [CMD].Please check the response packet.
   2. The response waiting time before sending the first data frame after CALL START is HEAD FRAME Time (80ms) plus the time of DPMR_PREAMBLE_LENGTH if set. The response time of the following frames is 80ms.

4. If the length of valid data is shorter than the length needed (DATA1 36 Bytes or DATA2 20Bytes),can I send the valid data only without fill the needed length?

   NO. No matter the length of valid data is shorter or equal to the max length, we must fill the max length.

5. How I can ensure the receiver receive the Header Frame? How long can SCT3258 report the Header Frame and output audio according the DPMR protocol?

   If HT_CI_PM item is to be reported (configured with REPORT_FIELD), the SCT3258 will report HT_CI_PM after detecting the carrier which and receiving the Header Frame.
   The shortest report time is a HEAD FRAME Time (80ms) plus the time of DPMR_PREAMBLE_LENGTH and the time MCU processing time..

Confidential，试样
Internal Use Only

6. When I send DATA TYPE2, the receiver will promote the FEC is error?

   When the BER is low, please check if the field END_FRAME is set or not. If it is set, please make sure it is set before CALL_START.

7. Can I make SCT3258 enter CARRIER READY state before it receives the RF carrier?

   It is better to make SCT3258 enter CARRIER READY state after it receives the RF carrier. Because when SCT3258 is receiving, it has to estimate the DC. If the CARRIER_READY state is entered before carrier is receiver,, the prediction may be wrong, which lead to poor receiver performance.

Confidential，供样
Internal Use Only

Copy Rights © 2013, Wuxi Sicomm Communication Technologies, Inc